

06 Mar 2024

DOMAIN CONTROLLER

Topics

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- ❖ Big Picture
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 - Build Process
 - Run Process
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 - Run Process
- ❖ Architecture
- ❖ Build Tools/Scripts
- ❖ Verification Tools

Introduction

DC collabs with satellite sensors, to perform certain operations, which a standalone sensors performs by itself.

DC sits in middle of ego Vehicle, connected to single or multiple sensors, placed at various positions on car.

DC is responsible to execute tracker module, to get Objects from detection received from connected sensors.

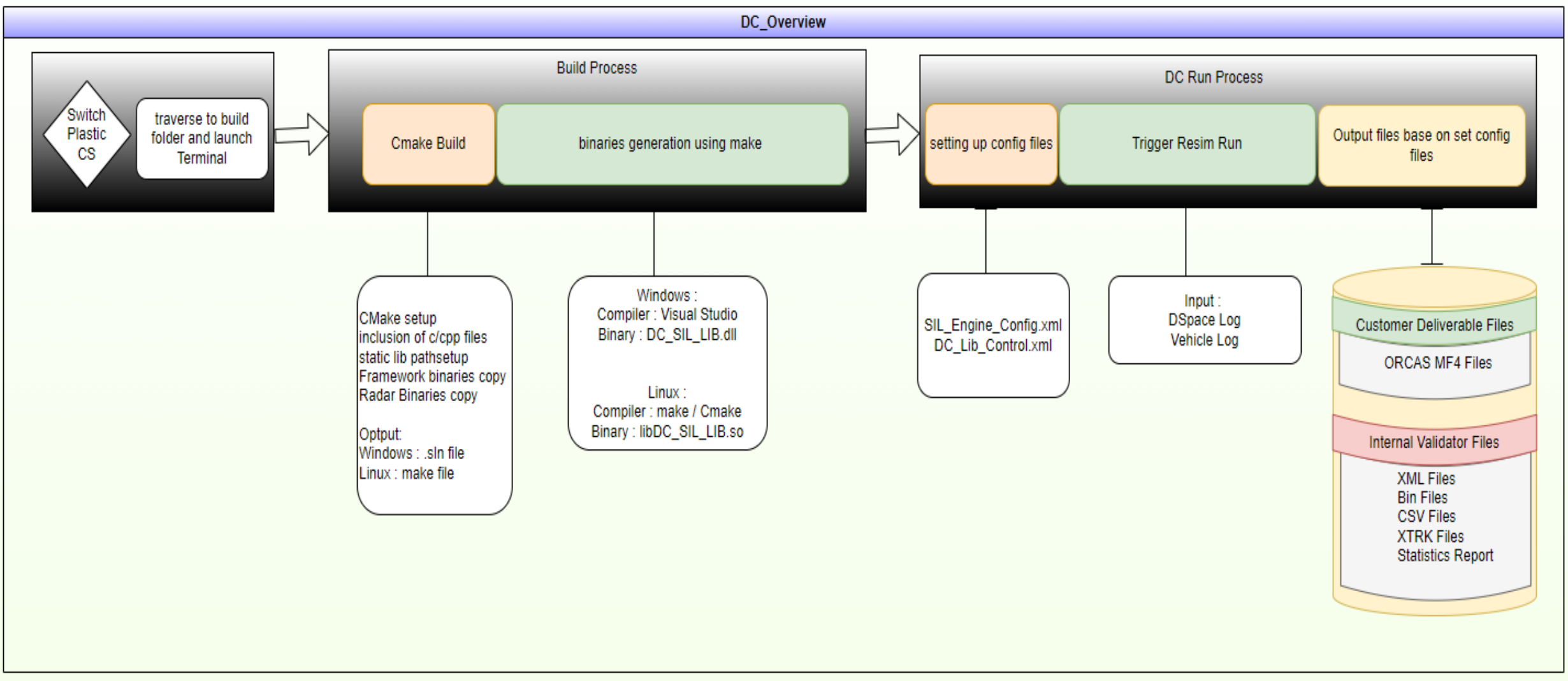
DC can be configured to execute SRR and MRR tracker.

SRR/MRR can be further configured based on tracker variants for individual Customer.

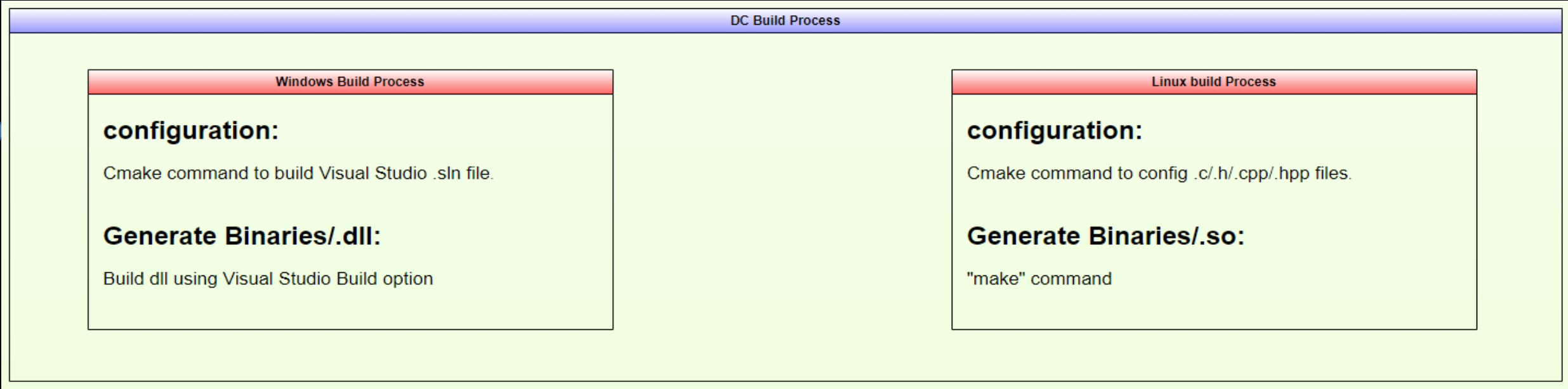
DC is responsible to run Feature Function Algorithms, as a part of ADAS (may differ based on customer)

DC is a RECU, takes input from one or more connected sensors, run algorithms and help to assist ego-vehicle, By communicating processed data to different ecu(s).

Big Picture : DC Overview



Big Picture : DC Build Process



Configuration Cmake Generic Commands

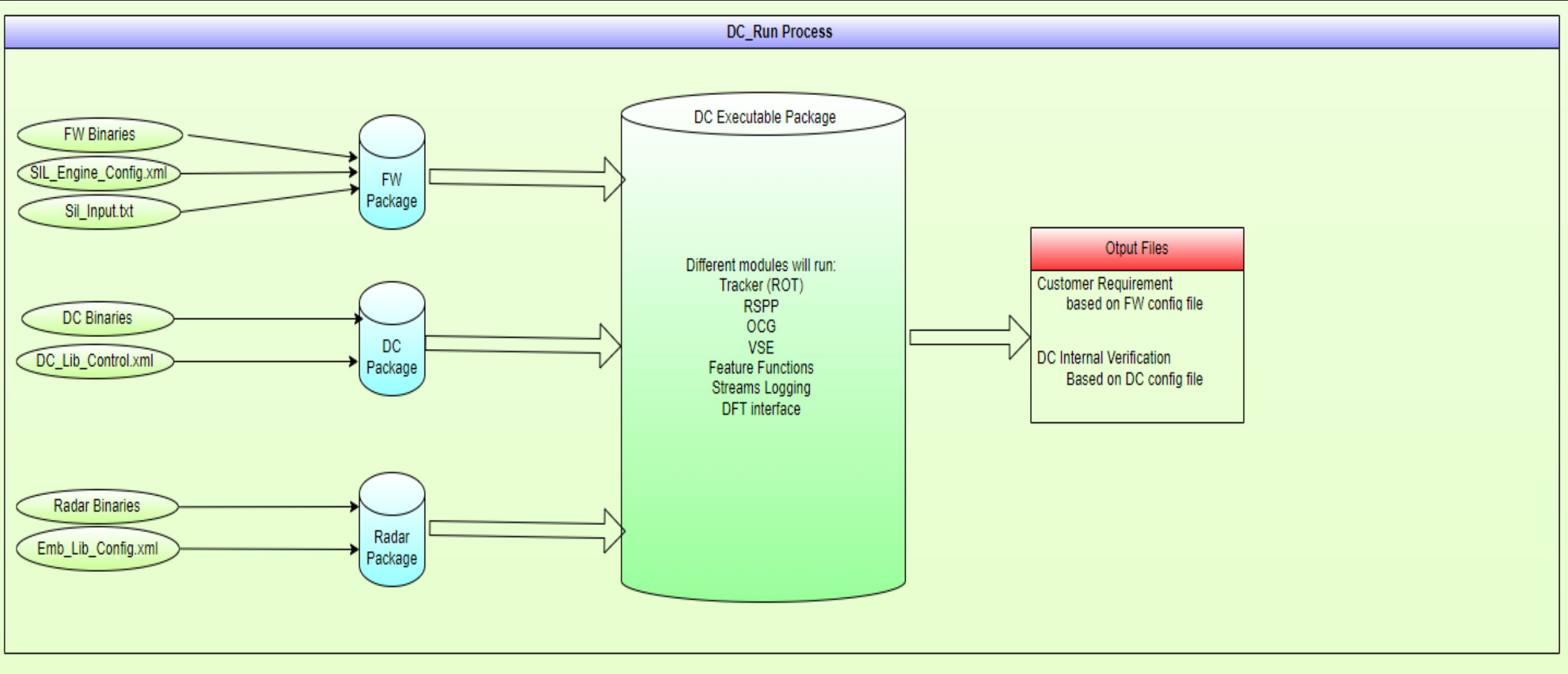
Windows : cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST="customer_name" -DTRACKER_TYPE="Tracker" -DVARIANT="customer variant" -G "Visual Studio 16 2019" ..
Linux : cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST="customer_name" -DTRACKER_TYPE="Tracker" -DVARIANT="customer variant" ..

Example:
Windows :
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=BMWSP25 -DTRACKER_TYPE=SRR -DVARIANT=BMW_SP25_L2 -G "Visual Studio 16 2019" ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=BMWSP25 -DTRACKER_TYPE=MRR -DVARIANT=BMW_SP25_L2 -G "Visual Studio 16 2019" ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=BMWSP25 -DTRACKER_TYPE=SRR -DVARIANT=BMW_SP25_L3 -G "Visual Studio 16 2019" ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=STLAMY24 -DTRACKER_TYPE=SRR -DVARIANT=STLA_SCALE1 -G "Visual Studio 16 2019" ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST= STLAMY24 -DTRACKER_TYPE=MRR -DVARIANT= STLA_SCALE1 -G "Visual Studio 16 2019" ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST= STLAMY24 -DTRACKER_TYPE=SRR -DVARIANT= STLA_SCALE1 -G "Visual Studio 16 2019" ..

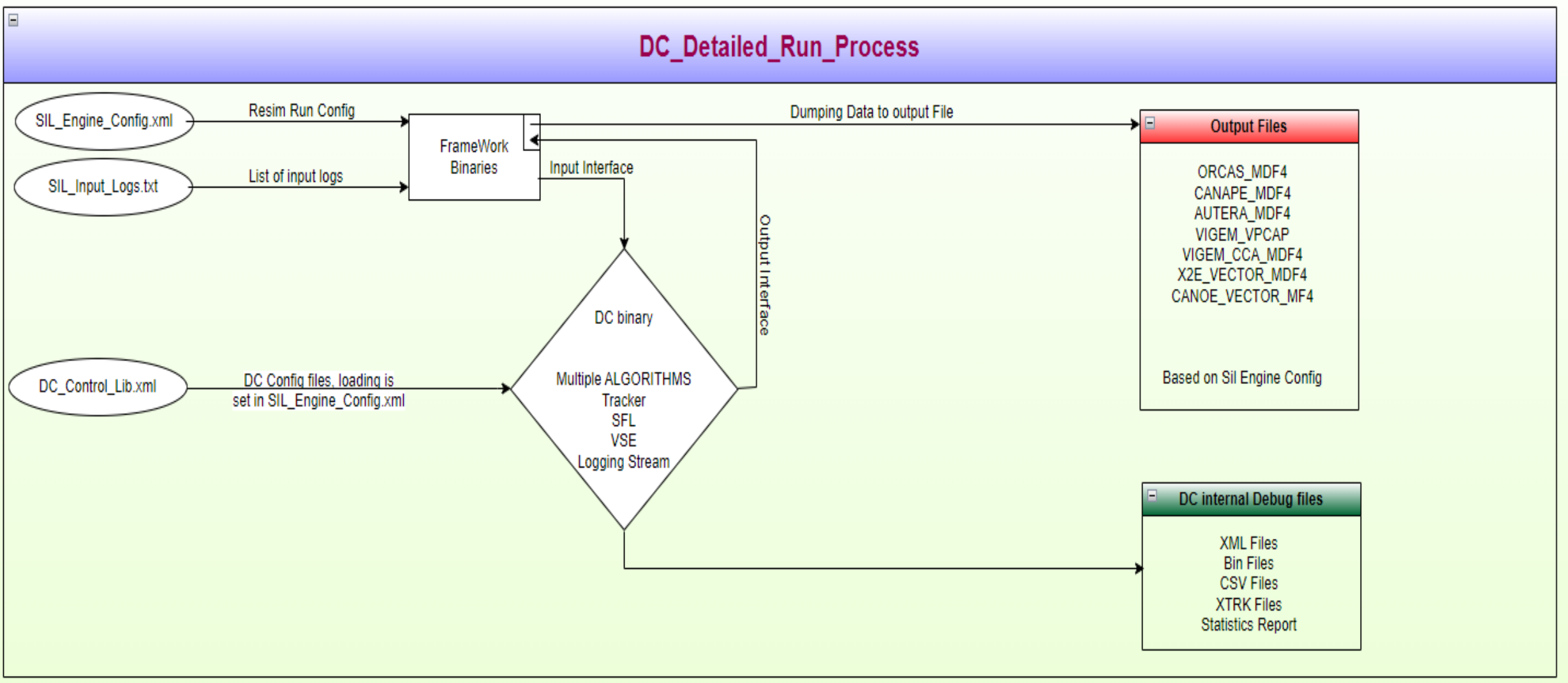
Linux :
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=BMWSP25 -DTRACKER_TYPE=SRR -DVARIANT=BMW_SP25_L2 ..
cmake -DCMAKE_BUILD_TYPE=DEBUG -DCUST=BMWSP25 -DTRACKER_TYPE=MRR -DVARIANT=BMW_SP25_L2 ..

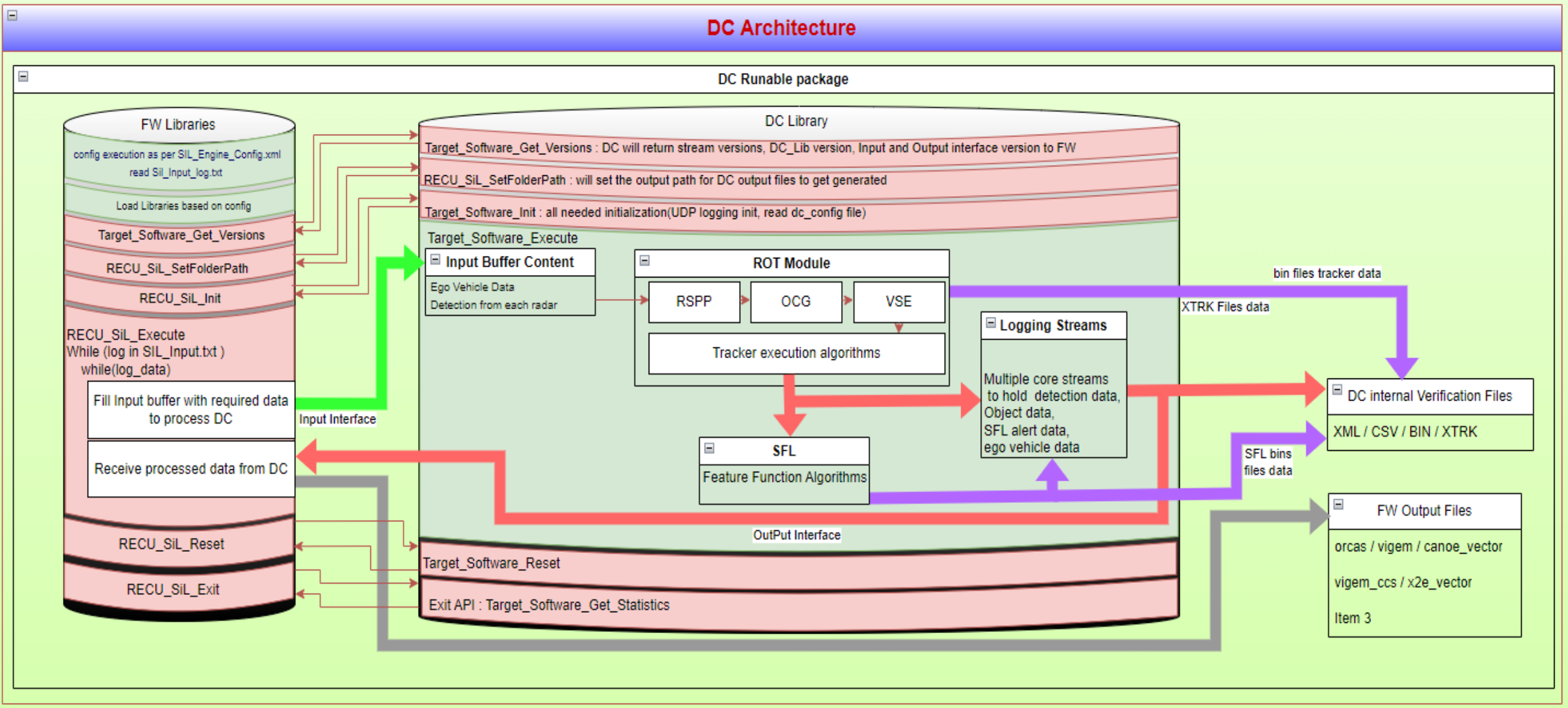


Big Picture : DC Run Process



Detail Picture : DC Run Process





Build and Validation Tools/Scripts:

Build and Run Tool/Script

<u>DC_Studio</u>	:	Build DC and Framework
DC_Automate	:	Script to builds multiple DC cust, with multiple possible combinations
RET	:	Script to execute RESIM for n number of log set, single or multiple times

Verification Tools/software

ASPE Tool	:	XTRK files
ORCAS Tool	:	Orcas MF4 Files
MATE Tool	:	Bin Files
Beyond Compare	:	XML Files

Build Tool : Cmake and Binaries build

VS-2019DCIpNextBMW_L2MRRGenerate SolutionDEBUGBRC

VS_version :ProjectCustomerVariantTrackerActionBuild Type

Path : /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/buildBrowse

ExecuteRelease Mode

Output WindowError Window

-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
[INFO] : checking OS configuration
[INFO] : 64 bit OS
F360 Tracker: Used variant -> D
F360 SW tag: 0x0000000
Entered GNU
RSPP: Used variant -> D
RSPP: Used variant -> D
RSPP SW tag: 0x0000000
[INFO] : SIL_SOURCE CORE linking started for BMWSP25
[INFO] : SIL_SOURCE CUSTOMER linking started for BMWSP25
CMake Deprecation Warning at /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/SIL_INT/DC_SIL_SOURCE/SIL_CORE/AS_BIN_WRAPPER/CMa...
Compatibility with CMake < 2.8.12 will be removed from a future version of CMake.

Update the VERSION argument <min> value or use a ...<max> suffix to tell CMake that the project does not need compatibility with older versions.

-- Performing Test COMPILER_HAS_HIDDEN_VISIBILITY
-- Performing Test COMPILER_HAS_HIDDEN_VISIBILITY - Success
-- Performing Test COMPILER_HAS_HIDDEN_INLINE_VISIBILITY
-- Performing Test COMPILER_HAS_HIDDEN_INLINE_VISIBILITY - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
[INFO] : Synchronizing FrameWork Binaries with x-link
[INFO] : Synchronizing Radar Binaries with x-link
-- Configuring done
-- Generating done
-- Build files have been written to: /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/build

0%APTIV15:29:25Help: mandeep.singh1@aptiv.com [dlcse7]version : 3.2.6

VS-2019DCIpNextBMW_L2MRRGenerate SolutionDEBUGBRC

VS_version :ProjectCustomerVariantTrackerActionBuild Type

Path : /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/buildBrowse

ExecuteRelease Mode

Output WindowError Window

LIB/CMaFiles/DC_SIL_LIB.dir/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/Application/F360Tracker/Wrappers/Common/source/f360_math_func.cpp.o -MF CMakeFiles/DC_SIL_LIB.dir/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/Application/F360Tracker/Wrappers/Common/source/f360_math_func.cpp.o.d -o CMakeFiles/DC_SIL_LIB.dir/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/Application/F360Tracker/Wrappers/Common/source/f360_math_func.cpp.o -c /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/Application/F360Tracker/Wrappers/Common/source/f360_math_func.cpp
ccplus: warning: command line option '-fgnu89-inline' is valid for C/ObjC but not for C++
[100%] Linking CXX shared library /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libDC_SIL_LIB.so
cd /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/build/DC_SIL_LIB && /usr/bin/cmake -E cmake_link_script CMakeFiles/DC_SIL_LIB.dir/link.txt --verbose=1
/usr/bin/c++ -fPIC -g -shared -Wl,-soname,libDC_SIL_LIB.so -o /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libDC_SIL_LIB.so CMakeFiles/DC_SIL_LIB.dir/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/SIL_INT/APTV_RECU_RESIM/ApplicationProjects/RECU_RESIM/RECU_SIL_LIB/RECU_SIL_fw_wrapper.cpp.o CMakeFiles/DC_SIL_LIB.dir/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/Application/F360Tracker/Wrappers/Common/source/f360_math_func.cpp.o -L/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/./Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug -Wl,-rpath,/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/./Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libCore_Interface.a ../FeatureFunctions/CED/libCED.a ../FeatureFunctions/LCDA/libLCDA.a ../FeatureFunctions/CTA/libCTA.a ../FeatureFunctions/PT/libPT.a ../FeatureFunctions/TA/libTA.a ../FeatureFunctions/RECU/libRECU.a ../FeatureFunctions/SCW/libSCW.a ../AS-bin-writer-lib/libAS-bin-writer-lib.a ../RECU_RESIM/pugixml PROJ/libpugixml.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libecu_sync.a ../F360Tracker/libbsp_25_tracker_wrapper.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libRECU_RESIM.a ../F360Tracker/Shared_Toolbox/Shared_Toolbox/libShared_Toolbox.a ../FeatureFunctions/Feature_Building_Kit/libFeature_Building_Kit.a ../FeatureFunctions/LTB/libLTB.a ../F360Tracker/VSE_Core/libVSE_Core.a ../F360Tracker/VRU_Classifier/libVRU_Classifier.a ../F360Tracker/Output_Adaptation/libOutput_Adaptation.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libMdf_Log_Ecu.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libCore_Interface.a ../FeatureFunctions/CED/libCED.a ../FeatureFunctions/LCDA/libLCDA.a ../FeatureFunctions/CTA/libCTA.a ../FeatureFunctions/PT/libPT.a ../FeatureFunctions/TA/libTA.a ../FeatureFunctions/RECU/libRECU.a ../FeatureFunctions/SCW/libSCW.a ../F360Tracker/libbsp_25_tracker_wrapper.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libMdf_Log_Ecu.a ../F360Tracker/rssp/librsp_variant_D.a ../F360Tracker/ocg/core/liboccupancy_grid.a ../F360Tracker/ocg/core/underdrivability/libocg_underdrivability.a ../F360Tracker/ocg/core/common/libocg_common.a ../AS-bin-writer-lib/libAS-bin-writer-lib.a ../RECU_RESIM/pugixml PROJ/libpugixml.a /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/Sil_Executable/RECU_dll/BMW_SP25_L2/MRR/Debug/libecu_sync.a -pthread ../FeatureFunctions/CalibrationTool/c_src/libCT_CALTOOL.a ../F360Tracker/F360TrackerLib/libf360_variant_D.a ../F360Tracker/Shared_Toolbox/Shared_Toolbox/libShared_Toolbox.a
make[2]: Leaving directory '/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/build'
[100%] Built target DC_SIL_LIB
make[1]: Leaving directory '/home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/build'
/usr/bin/cmake -E cmake_progress_start /home/dl31tmx2/wkspaces/Working_Repos/IpNext_Simulation/DC_SIL_Library/CMake/build/CMakeFiles 0

100%APTIV15:28:34Help: mandeep.singh1@aptiv.com [dlcse7]version : 3.2.6

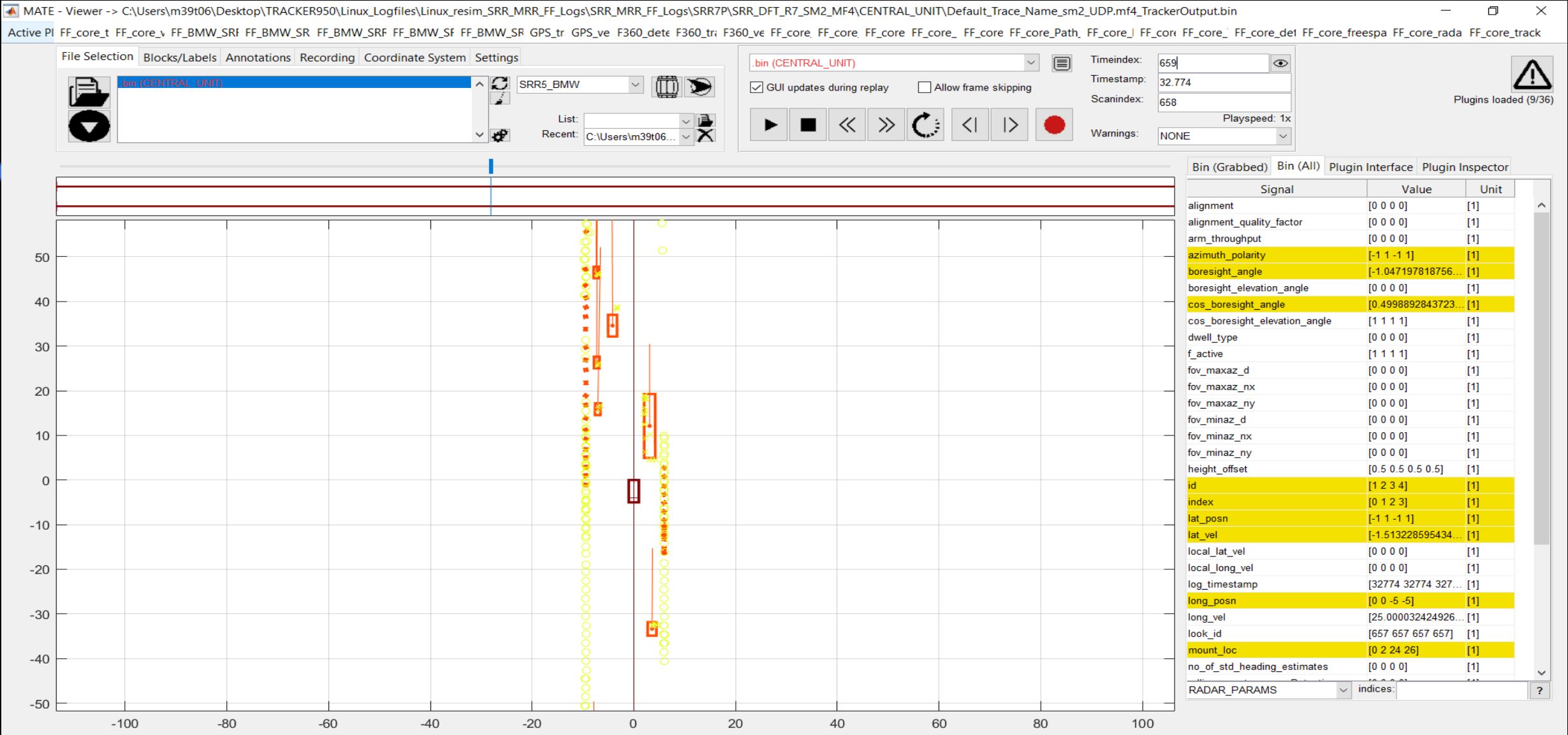
Domain Controller | March-6 2024 | Aptiv Confidential


```
*****
*                                     *
*      APTIV                         *
*                                     *
*****
*                                     *
*                               AutoDC *
*                                     *
*****
* version: 2.0                      Help : mandeep.singh1@aptiv.com *
*****
*****

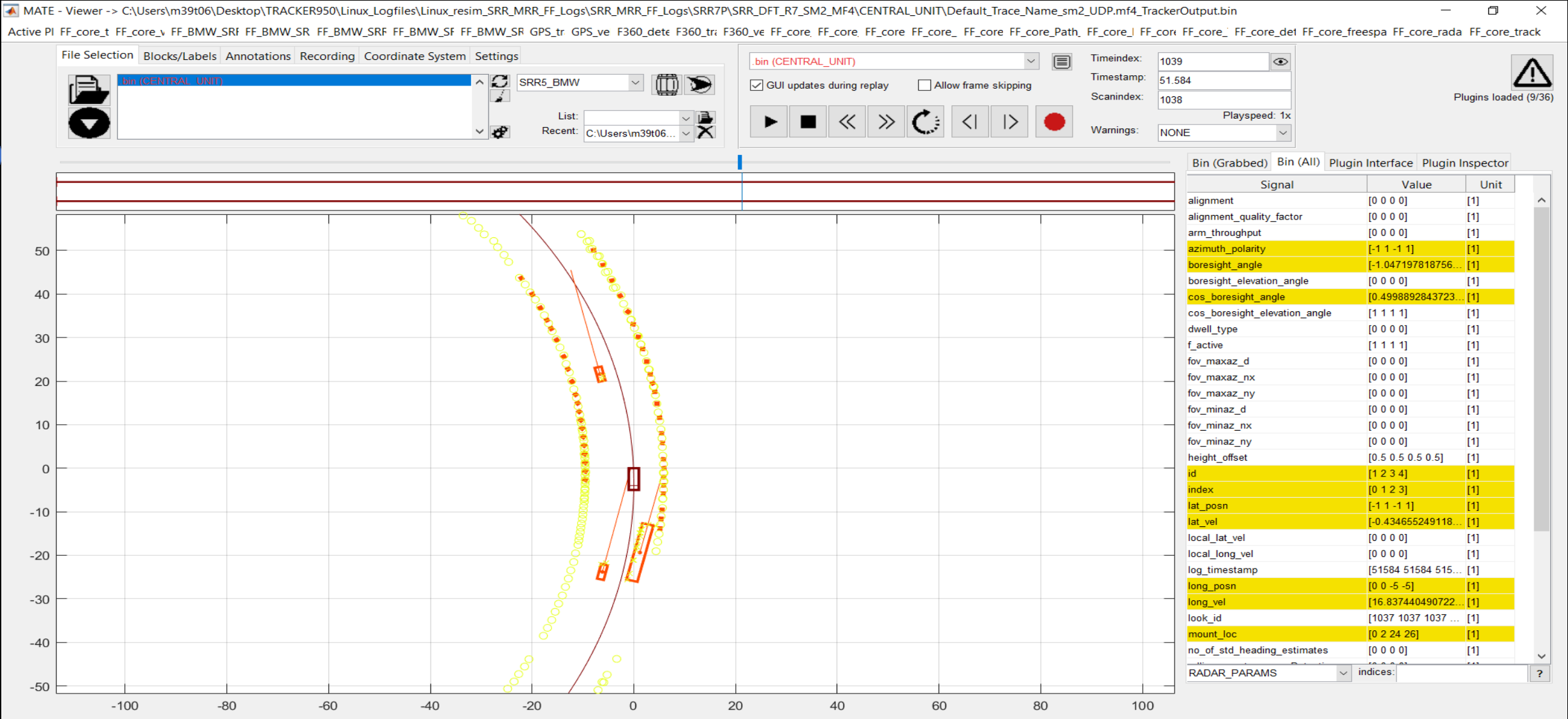
Select task to perform by entering the *number* before the desired choice.
1) All_Customers
2) IpNext_Debug
3) IpNext_Release
4) IFV600_Debug
5) IFV600_Release
6) Debug_Mode
7) Release_Mode
8) exit
```


Executing Loop Count:1					
Log Set	Execution	Log Run Status	ReSim Time	Scan Drop	Remark
Set_SRR7P	Completed	Successful	61 sec	1 drops	Resimulation Operation Completed...
Set_SRR5P	Completed	Successful	152 sec	2 drops	Resimulation Operation Completed...
Set_SRR6P	Completed	Successful	195 sec	2 drops	Resimulation Operation Completed...

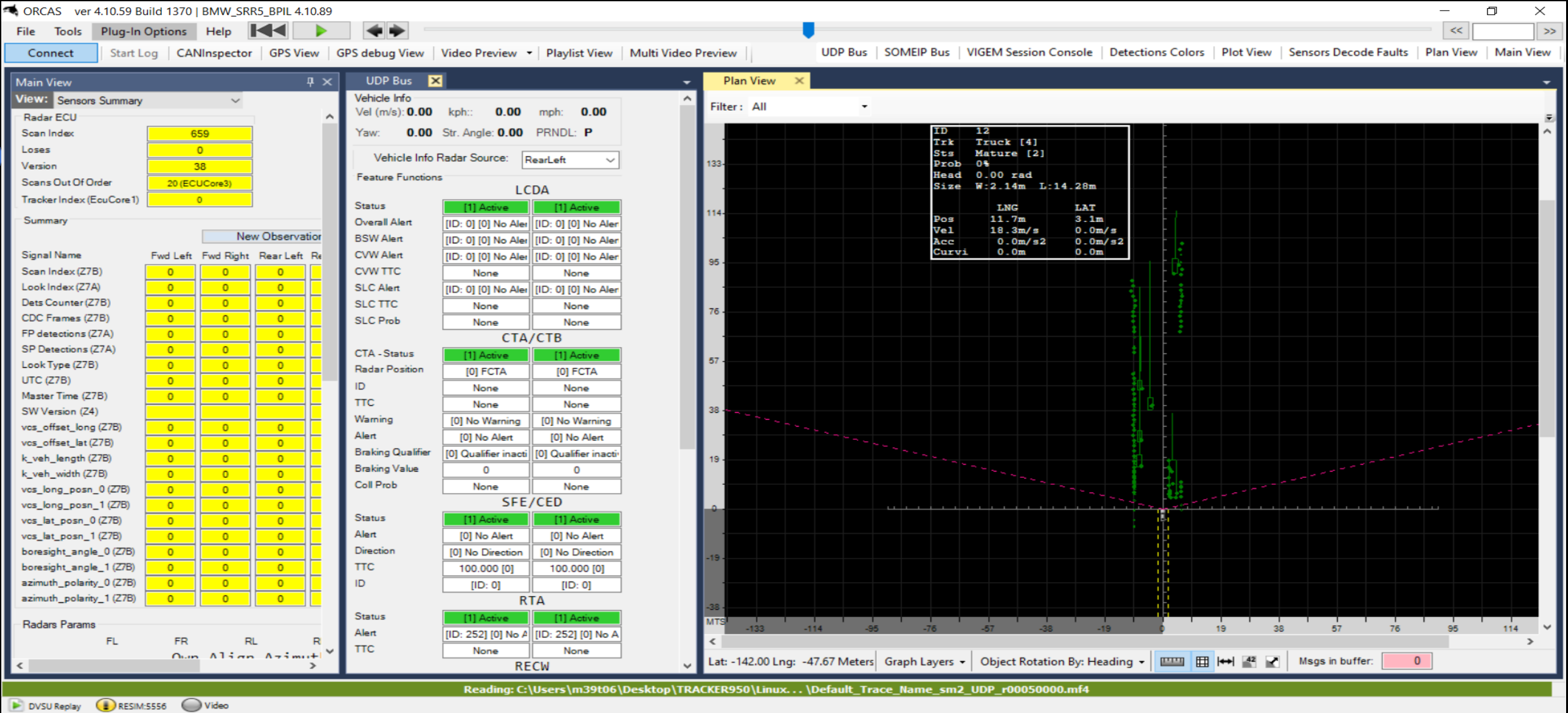
Verification Tool : MATE Tool



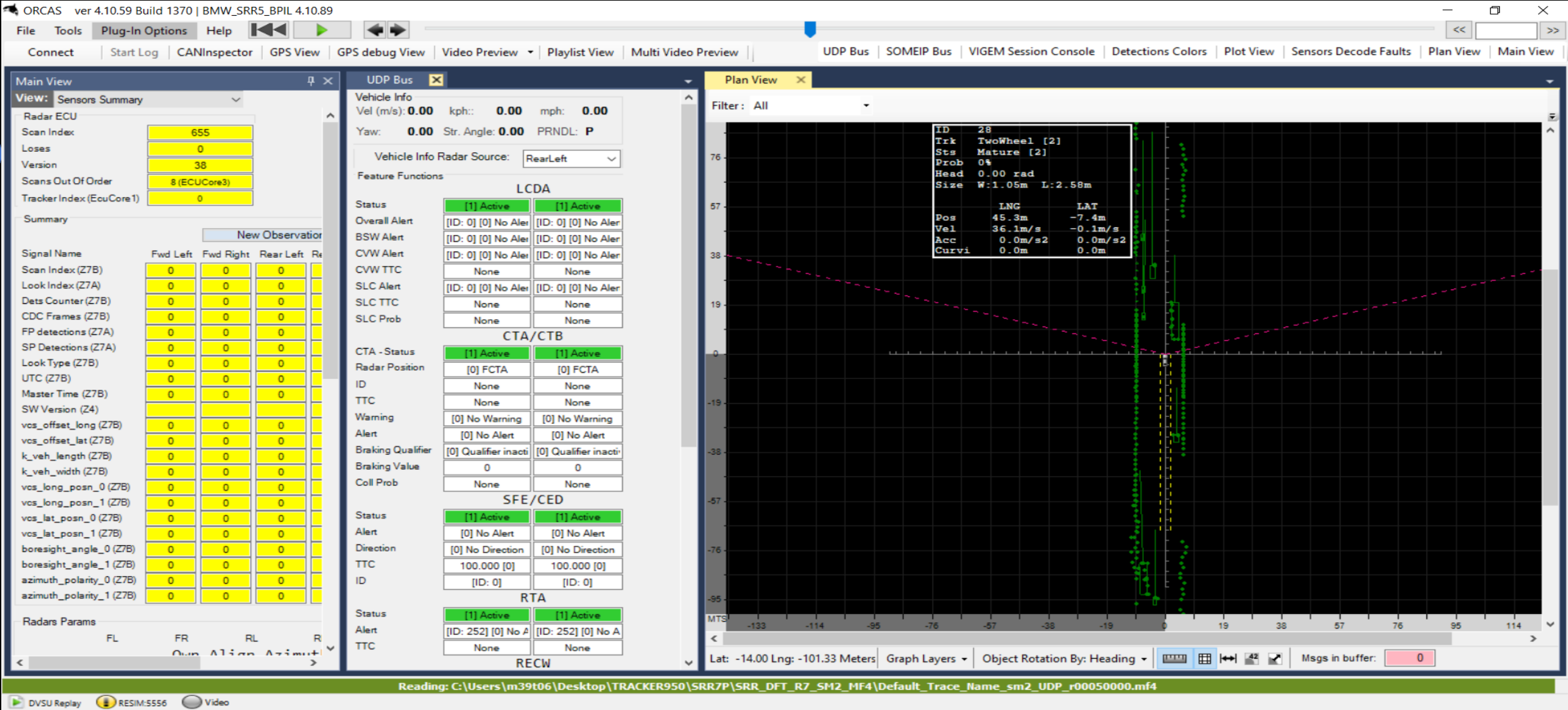
Verification Tool : MATE Tool



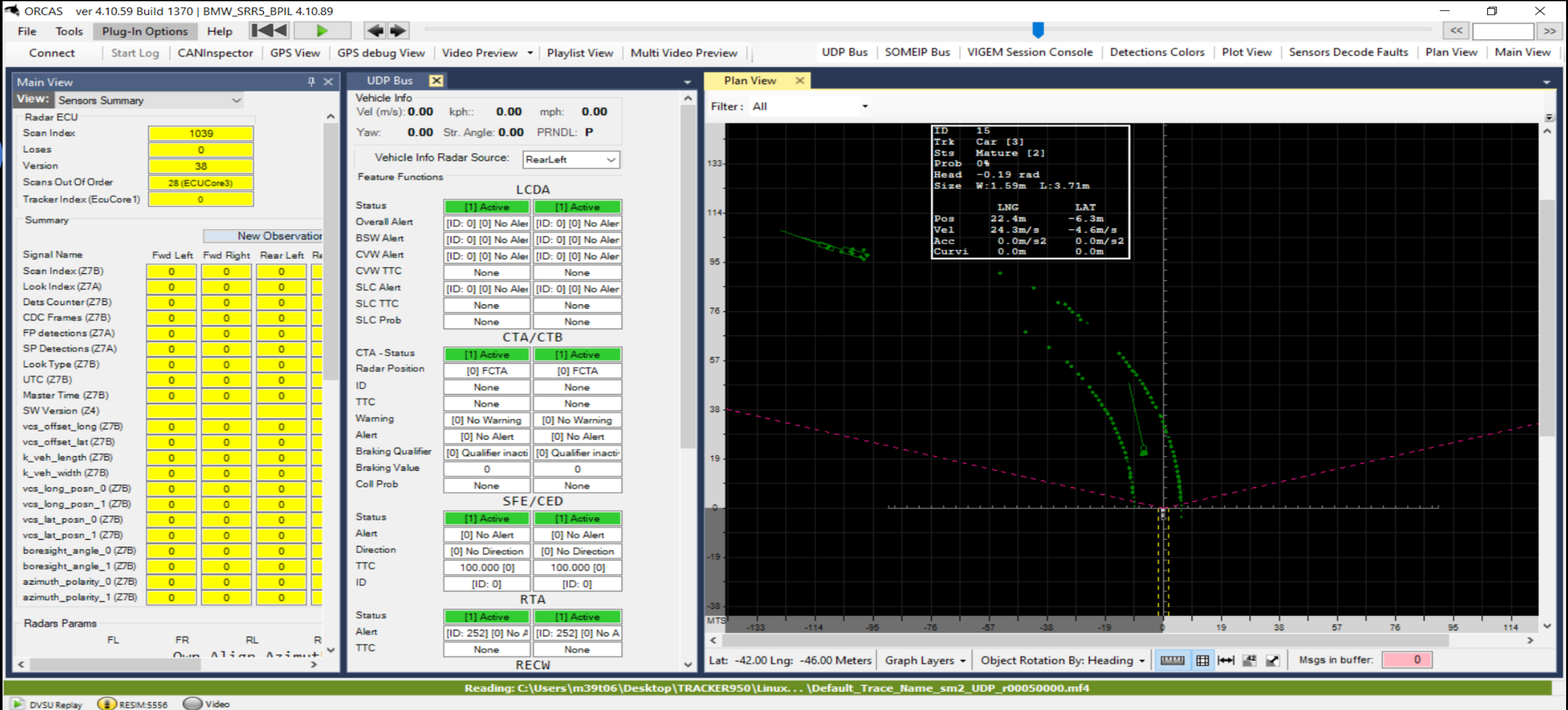
Verification Tool : ORCAS Tool



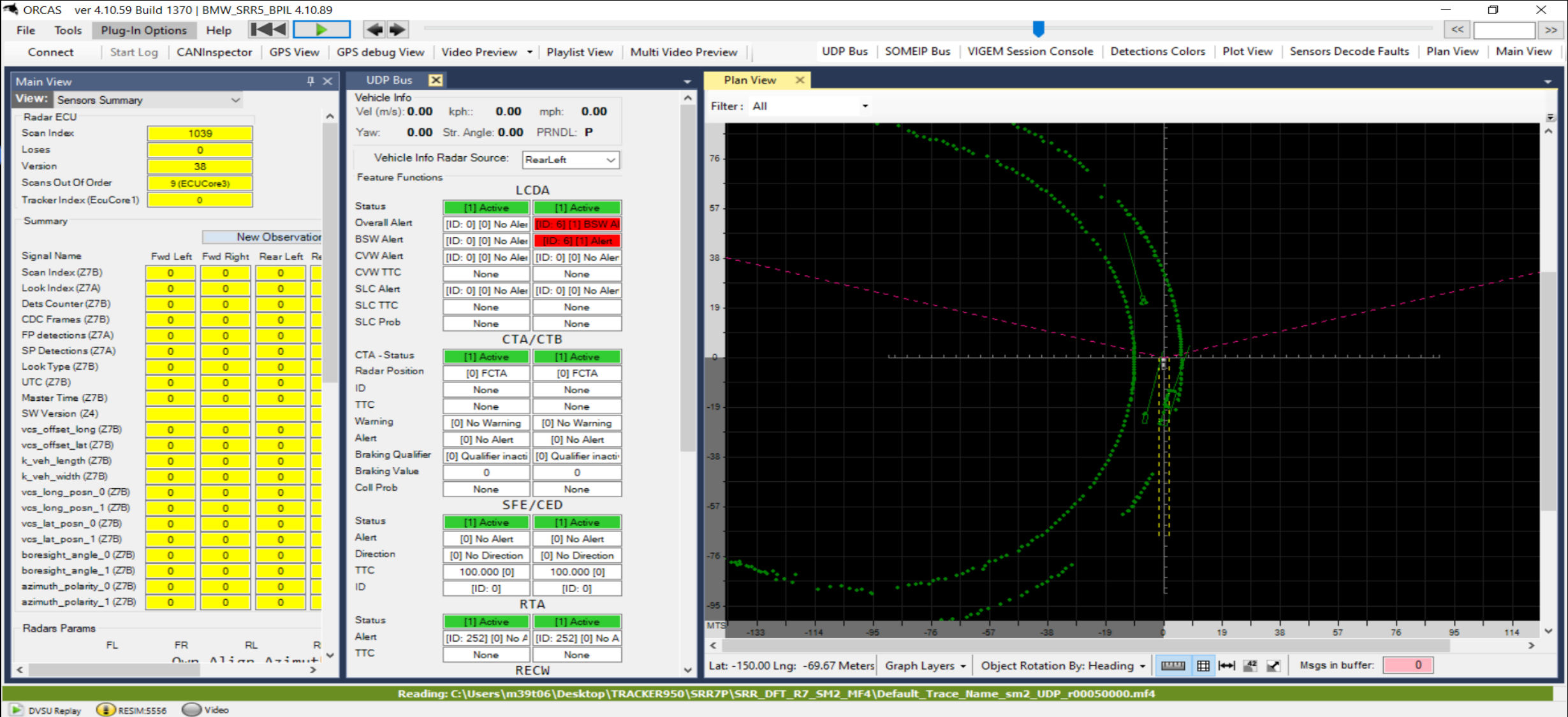
Verification Tool : ORCAS Tool



Verification Tool : ORCAS Tool



Verification Tool : ORCAS Tool



Verification Tool : ASPE Tool

ASPE GUI

Settings

LoadDataPanel

C:\Users\m39t06\Desktop\Debug\SRR_10_0_0\XTRK\SRR_Default_Trace_Name_sm2_UDP.xtrk

Open

subfolder

type

1 XTRK F360 .xtrk data

PlaybackPanel

Play < > R S REC

Scan index 617

Playback speed x 1.0

Timestamp

ToolboxPanel

DataViewer

DataSource SRR_Default_Trace_Name_sr

Data set internal_objects

Only current scan ☐

raw_signals signals signals_info

accuracy_length	accuracy_width	average_r
0.0	0.0	1.9674158
0.0	0.0	2.0267532
3.5	3.5	3.0509124
3.5	3.5	2.4677405
3.5	3.5	3.206981

Drawers settings

Selection table view

Verification Tool : ASPE Tool

ASPE GUI

Settings

LoadDataPanel

C:\Users\m39106\Desktop\Debug\SRR_10_0_0\XTRK\SRR_Default_Trace_Name_sm2_UDP.xtrk

Open

subfolder

1 XTRK

F360 .xtrk data

type

PlaybackPanel

Play

<

>

R

S

REC

Playback speed

x 1.0

Scan index

980

Timestamp

ToolboxPanel

DataViewer

DataSource

SRR_Default_Trace_Name_sr

Data set

internal_objects

Only current scan

☐

raw_signals

signals

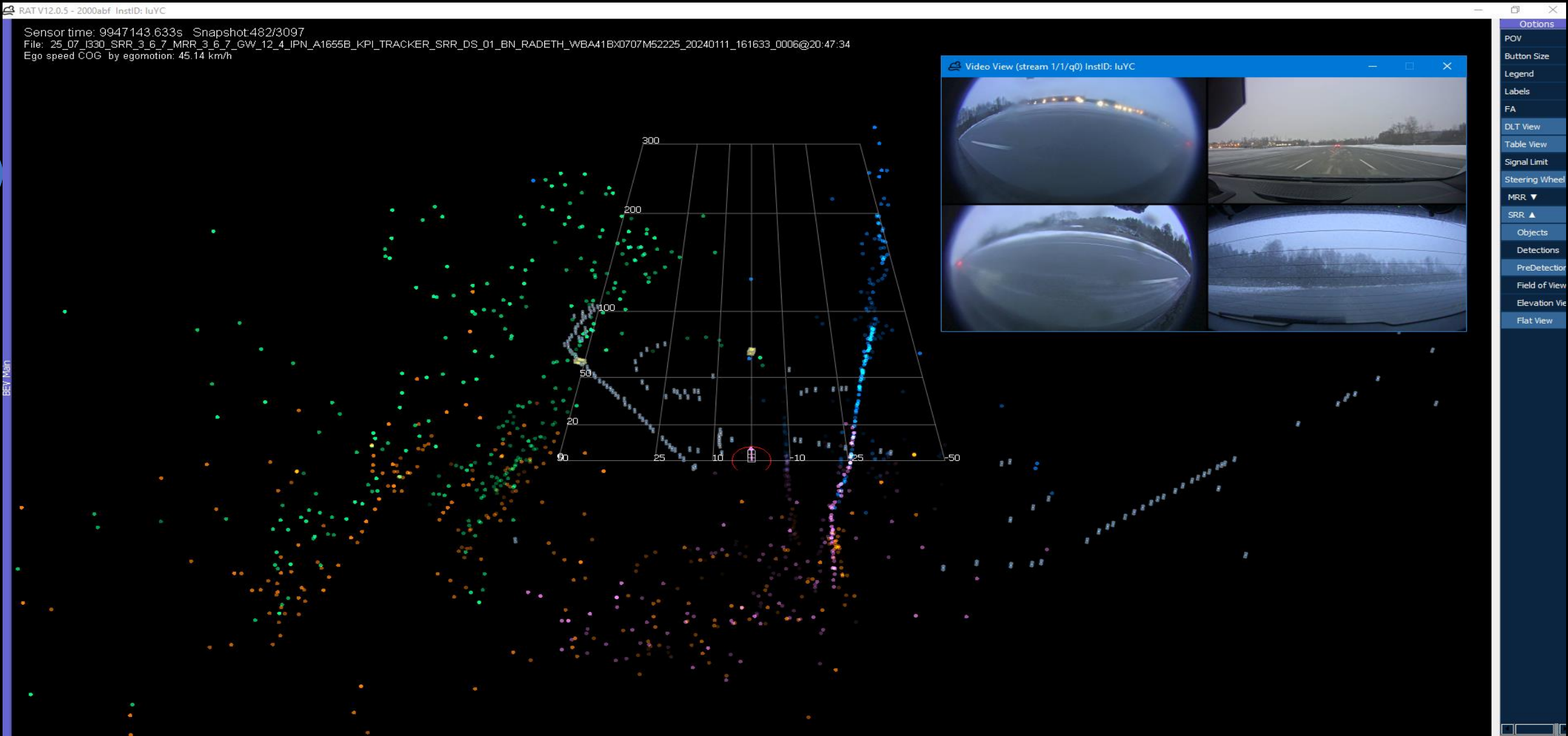
signals_info

accuracy_length	accuracy_width	average_r
0.0	0.0	1.9674158
0.0	0.0	2.0267532
3.5	3.5	3.0509124
3.5	3.5	2.4677405
3.5	3.5	3.206981

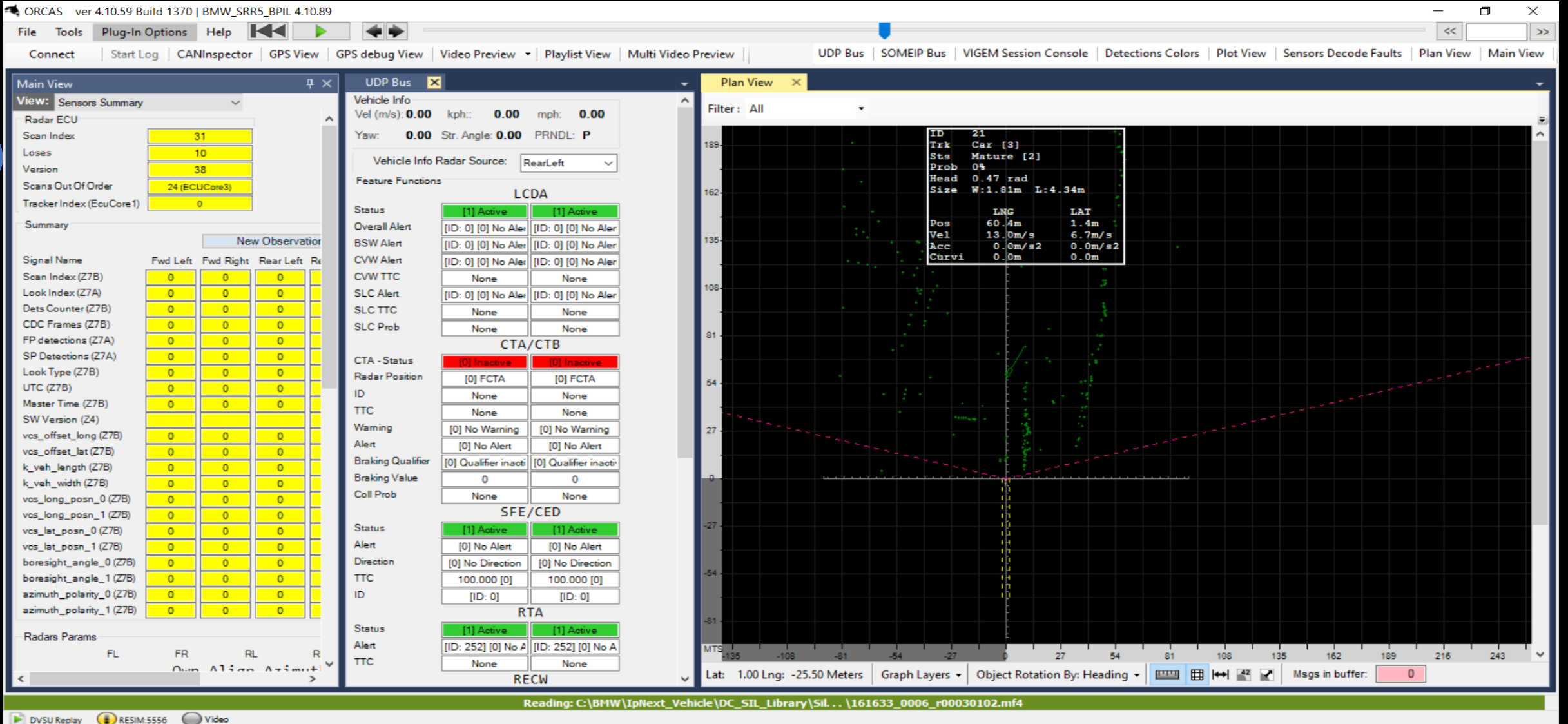
Drawers settings

Selection table view

Verification Tool : RAT Tool



Verification Tool : ORCAS Scenario wrt RAT Tool



Questions ???

Thank U